

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech 3rd Semester Mid-Semester Examination, September 2024
Subject: Mathematics - III/1 (MA 2101)
(CST, EE, IT and ME)

Time : 2 hours

Full Marks : 30

Answer any FIVE questions.

2023 CSB046

1. (a) Prove that $P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B)$, for any two events A and B . [2]
- (b) Show that the probability that exactly one of the events A and B occurs is $P(A) + P(B) - 2P(A \cap B)$. [2]
- (c) Two dice are thrown simultaneously. What is the probability that the total sum of the points on the dice will be 6? 5/36. [2]
2. (a) State the Baye's theorem. [2]
- (b) There are three identical urns containing white and black balls. The first urn contains 2 white and 3 black balls, the second urn contains 3 white and 5 black balls and the third urn contains 5 white and 2 black balls. An urn is chosen at random, and a ball is drawn from it. If the ball drawn is white, what is the probability that the second urn is chosen? 35/139 [4]
3. (a) A random variable X has the following probability mass function:

X	0	1	2	3	4
$P(X = x_i)$	0	$5k$	$3k$	k	k

Determine the value of k . 1/10

[2]

- (b) Find the constant 'C' so that the function

$$f(x) = \begin{cases} Cx(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

is a probability density function. Find the corresponding distribution function. [1+3]

4. (a) A random variable X takes the values $x_i = \frac{2^i}{i}, i = 1, 2, \dots, \infty$; with probabilities $p_i = \frac{1}{2^i}$. Then find the expectation of X , if it exists. [2]
- (b) If the probability density function $f(x)$ of a random variable X is defined by $f(x) = \frac{1}{2}e^{-|x|}, -\infty < x < \infty$; then evaluate the variance of X . [4]

5. (a) Two players A and B play tennis game. The probabilities of winning a game are in the ratio $3 : 2$ respectively. Find the probability of winning of A at least two games out of four games played. [3]

(b) Calculate the mean and variance of Poisson distribution. [1+2]

6. Find all the basic solutions for the system of equations:

$$2x_1 + 6x_2 + 2x_3 + x_4 = 3$$

$$6x_1 + 4x_2 + 4x_3 + 6x_4 = 2.$$

$$(0, 1, 2, 0, 0).$$

Identify the basic feasible solutions among all the basic solutions. [5+1]

7. Show that $(1, 1, 1)$ is a feasible solution to the system of equations

$$x_1 + x_2 + 2x_3 = 4$$

$$2x_1 - x_2 + x_3 = 2.$$

Is this solution a basic feasible solution? If not, reduce this given feasible solution to a basic feasible solution. $(2, 2, 0)$ & $(0, 0, 2)$. [1+1+4]

Indian Institute of Engineering Science and Technology, Shibpur
B. Tech (CST), 3rd Semester, Mid-Semester Examination, September, 2024
Discrete Structures (CS2101)

Full Marks: 30

Time: 2 Hours

- Answer only three questions
- No answer to extra question will be evaluated.

203 CS8046

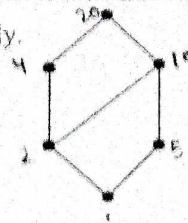
1. Choose the correct alternative. No explanation is needed for your answer.

(a) If $U = \{a, b, c, d, e, f, g, h\}$, $A = \{a, b, c, g\}$, $B = \{d, e, f, g\}$ then $\overline{A \cap B}$ is
(i) $\{g, h\}$ (ii) $\{h\}$ (iii) $\{d, e, f\}$ (iv) Φ

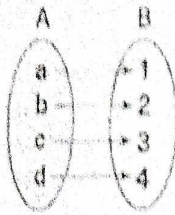
(b) $\neg(p \leftrightarrow q)$ is same as
(i) $p \vee \neg q$ (ii) $\neg p \vee q$ (iii) $p \wedge \neg q$ (iv) ~~$p \vee \neg q$~~ ^{None} where $p, q \in \{0, 1\}$

(c) If $(D_n, |)$ is a poset, D_n : divisors of n then what is the value of n for the following Hasse diagram? Here, labels of the diagram are not provided intentionally.

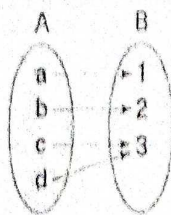
(i) 16 (ii) 12 (iii) 20 (iv) 10



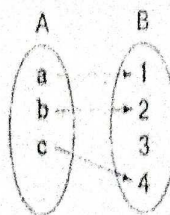
(d) Which of the following is injective function?



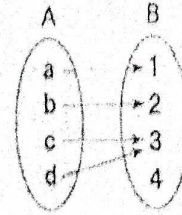
$f_1 \checkmark$



$f_2 \times$



$f_3 \checkmark$



$f_4 \times$

(i) Only f_1 (ii) Both f_1 and f_3 (iii) Only f_1 (iv) Only f_3

(e) In a relation $R = \{(p, 1), (q, 3), (r, 2), (s, 4)\}$, then $a \notin R^{-1}$ is

(i) $(4, s)$ (ii) $(3, q)$ (iii) $(2, r)$ (iv) $(3, p)$

(f) If $f(x) = 2x + 3$, $g(x) = x^2 - 1$ then $(f \circ g)(x)$ is

(i) $2(x^2 - 1) + 3$ (ii) $2(x^2 - 1) - 3$ (iii) $(x^2 - 1) + 3$ (iv) $2x^2 - x + 3$

(g) Which of the following statements about the function $f(x) = x^2 - 4x + 3$ is true?

$\times$ (i) The function has no real roots.

(ii) The function is increasing for all values of x .

$\bullet$ (iii) The function can be factored into $(x-1)(x-3)$.

$\times$ (iv) The function is a linear function.

P.T.O

(b) Which of the following statements is a tautology?

(i) $(P \wedge Q) \vee \neg (P \vee Q)$

(ii) $P \vee \neg P$

(iii) $(P \rightarrow Q) \wedge (Q \rightarrow R)$

(iv) $P \wedge (Q \vee \neg P)$

(c) To prove the statement "If x is an odd integer, then x^2 is odd," which of the following is a correct direct proof?

(i) Assume x is odd, so $x = 2k + 1$ for some integer k . Then $x^2 = (2k + 1)^2 = 4k^2 + 4k + 2$ which is odd. So x^2 is odd.

(ii) Assume x is odd, so $x = 2k + 1$ for some integer k . Then $x^2 = (2k + 1)^2 = 2(2k + 1)$ which is odd. So x^2 is odd.

(iii) Assume x is odd, so $x = 2k + 1$ for some integer k . Then $x^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ which is form of $x = 2m + 1$ where m is an integer. So x^2 is odd.

(iv) Assume x is odd, so $x = 2k + 1$ for some integer k . Then $x^2 = (2k + 1)^2 = 4k^2 + 4k + 1$ which is even. So x^2 is odd.

(d) Given a set S where $|S| = 12$, how many subsets of S have exactly 3 elements?

(i) 220 (ii) 792 (iii) 924 (iv) 120

(10 × 1 = 10)

2. (a) Answer in one sentence only

(i) What is the proof technique where you assume the negation of the statement you want to prove and derive at something opposite to your assumption? **Contradiction**

(ii) In set theory, what is the term for the set of all subsets of a given set A ? **Power set**

(iii) In a partially ordered set (poset), what term describes the maximum element that is less than or equal to every element in the set? **least**

(iv) In a relation, what is it called when if a is related to b and b is related to c , then a is related to c ? **Transitive**

(v) What is the term for a function where every element in the codomain is mapped by at least one element in the domain? **Onto**

(b) Clearly write True or False for the following statements. No justification is needed for your answer.

(i) If $x \in [2, 3]$ and $y \in (2, 3)$ are $2 \leq x \leq 3$ and $2 < y < 3$ respectively, then $[1, 2) \cup [2, 3] = [2, 3]$. **F**

(ii) $\neg(p \rightarrow q) = p \wedge \neg q$ **T**

(iii) If $x, y \in \{1, 3, 5\}$, $R = \{(x, y) | x \leq y\}$ then R is reflexive. **T**

(iv) If $B = \{\{1, 2\}, 3, \Phi\}$ then $|B| = 3$ **T**

(v) "If P is a lady, then P is a human being"

In this statement " P is a lady" is NECESSARY condition for " P is a human being" and " P is a human being" is SUFFICIENT condition for " P is a lady". **F**

(5 + 5 = 10)

P.T.O

3. (a) Citing an example, compute $A \cup (A \cap B)$ where $A, B \subseteq N$

(b) Using membership table check if $A \cup (A \cap B) = A$

(c) Find $\bigcap_{i=1}^n A_i$ where $A_i = \{x \mid -i \leq x \leq i, \forall x \in \mathbb{Z}\}$, $n, i \in \mathbb{N}$

(d) Using Venn diagram show $\bigcap_{i=1}^n A_i$ for $n \leq 3$

$$(4 \times 2 \frac{1}{2} = 10)$$

4. Prove the following using the specified proof strategy

(a) The sum of two rational numbers is a rational number (direct proof).

(b) If $\bar{B} \subseteq \bar{A}$ then $A \subseteq B$ (method of contradiction).

(c) $n^2 + 5$ is odd, then $n \in \mathbb{Z}$ is even (method of contraposition).

(d) $\min(a, \min(b, c)) = \min(\min(a, b), c)$ (proof by cases).

$$(4 \times 2 \frac{1}{2} = 10)$$

5. (a) Consider the propositions, p : It is raining and q : The ground is wet. Represent the following compound propositions using logical connectives:

- The ground is wet if and only if it is raining.
- If it is not raining, then the ground is not wet.

(b) Define and provide an example of a tautology and a contradiction.

(c) Given the compound proposition $(p \wedge \neg r) \rightarrow (q \vee s)$, determine the inverse, converse, and contrapositive. Additionally, provide a truth table for the original proposition.

(d) Consider the following propositions:

- The sky is blue
- It is daytime

Explain whether the compound propositions that "During daytime the sky is blue" is satisfiable

$$(4 \times 2 \frac{1}{2} = 10)$$

1. (a) Determine the possible unknown base b of the following relation: $(\sqrt{22})_b = (6)_{10}$.
(b) Represent the following decimal numbers in BCD and perform the BCD addition. The two numbers are 28 and 39.
(c) Assume that in a combinatorial circuit with output F , the minterms can be given as below, where m denotes the minterms and d denotes the don't care conditionals.

$$f(a, b, c, d) = \sum m(0, 2, 5, 7, 9, 11) + \sum d(3, 8, 10, 12, 14)$$

$\overline{b}c, a\overline{c}d$ 3 $\overline{a}bd, a\overline{b}, \overline{b}\overline{d}$

Derive all the prime implicants and essential prime implicants using K-map. [2 + 3 + 5]

2. Design a combinational circuit with three inputs x, y, z , and has three outputs A, B , and C . When the binary input is 0, 1, 2 or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input. [10]
3. (a) Consider the combinational circuit given below. Derive the minimal sum-of-product expression for the output F . $1, 3, 5, 7$ $\overline{P}Q + \overline{P}Q + P\overline{Q}$

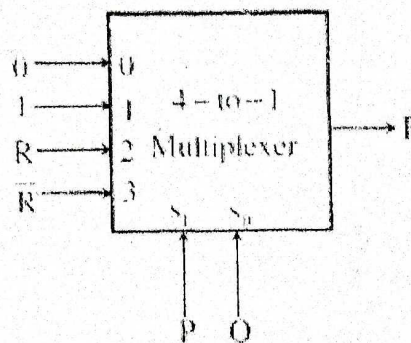


Figure 1: A combinational implemented using MUX.

- (b) Using four XOR gates and minimum number of full-adders, construct a 4-bit parallel adder/subtractor circuit. Use an input select variable S so that when $S = 0$, the circuit adds and when $S = 1$, the circuit subtracts. [6 + 4]
4. (a) Write the characteristic table and characteristic equation of $J - K$ flip-flop. Also implement the $J - K$ flip-flop using two-input NAND gates only.
(b) Is there any problem in $J - K$ flip-flop? If yes, what are the methods to solve that problem. [6 + 4]

Attempt all questions

- ✓ Suppose in a set of elements, only some are frequently searched, and the remaining are less frequently searched. The elements are arranged as a singly linked list.

You need to write a function named as `searchLinkedList(struct listNode * list, int elem, ...)` to search an item 'elem' in the list and return the search result. This function also should reorganize the linked list to improve the searching time for the subsequent calls (i.e., it takes less time to search for frequently searched elements).

Note: You may add additional field along with the data field in the `listNode` to solve this problem. You may change the prototype of the function as per your need.

(5)

- ✓ Answer any 5 from the following questions

- What is Big O notation? Why do we use Big O notation to compare algorithms?
- ✓ Calculate the address of a random element present at $(i, j)^{\text{th}}$ location in a 2-D array of size $M \times N$, given the base address as α .
- c) Convert the postfix form of the following infix expression using a stack (no code, only pictorial representation): $D / (A - B) + F / (E + A * G) - K$
- ✓ d) What are the merits and demerits of a doubly linked list over singly linked list?
- + e) Explain shallow copy and deep copy.
- ✓ f) Explain why complexity of Binary Search is $O(\log n)$?
- ✓ g) Define a structure to represent a stack with a finite number of elements. However, multiple stacks can be of different sizes. Write a method to initialize the data elements of the stack.
- ✓ h) Write down the recursive subroutine for calculating the F_n for an input positive number n . The F_n is calculated by setting $F_0 = 0$, $F_1 = 1$ and using the recursive formula $F_n = n * F_{(n-1)} + F_{(n-2)}$. Finally, trace the function calls for $n = 4$.

(5x3 = 15)

8. Answer any 2 from the following questions

- a) Given a singly linked list, write down a function that prints all the alternate nodes of it. For example, if the given linked list is $6 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 6$ then your function should print 6 3 6, and if the given linked list is $4 \rightarrow 2 \rightarrow 1 \rightarrow 5$ then it prints 4 1.
- b) Let's assume that a set of integers is stored in an array (named `myArr`) in random order. Write a C function `deleteAnElement(int key, ...)` to delete an element 'key' from the array so that the order of the remaining elements is intact.
- c) Write a C function to confirm whether a given string is palindrome (*reads the same backward or forward*). You must use a stack to solve this problem. You don't have to write stack-related functions; only stack related function calls are sufficient.

(2x5 = 10)

Indian Institute of Engineering Science and Technology, Shibpur
CST Third Semester Midterm Examination, September 2024
Signals and Systems (CS-2104)

2023 CS8046

There are seven questions of six marks each.
Each question has two related parts with equal weightage.
Answer any five questions OR ten parts.

Full Marks: 30

Time: 120 minutes

1. (a) Briefly explain the concept of linear time invariant systems.
(b) Describe how the impulse response of such system can be obtained.
2. (a) What is the connection between the frequency variables used in Fourier transform, Laplace transform and Z transform?
(b) Explain the mapping of the stable region of complex s plane to the stable region of z domain.
3. A second order system is described as $\ddot{y}(t) + 5\dot{y}(t) + 6y(t) = 3x(t)$.
(a) Obtain the transfer function of the system.
(b) What is the response of the system for step input?
4. A second order discrete system is described by the difference equation:
 $y[n] - 0.5y[n-1] + 0.04y[n-2] = 3x[n]$.
(a) What are the poles of the system in z domain?
(b) Represent the system using three different cascaded block diagrams.
5. A periodic square wave signal with peak to peak amplitude of 2 volts and frequency 10 KHz is passed through an integrator that has an integration coefficient of $0.01(ms)^{-1}$.
(a) Sketch the output signal waveform in time domain for one cycle.
(b) Show how this integrator can be realized in continuous domain.
6. Provide schemes for the following with underlying mathematics:
(a) Obtaining initial value, final value and all values of a discrete sequence from the z transform
(b) Obtaining the frequency components of a square wave with very large time period.
7. Write short notes on the following:
(a) Energy content of a periodic signal (discrete and continuous) in time domain and frequency domain.
(b) Sampling of a continuous signal and its effects in frequency domain.